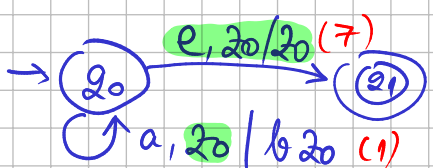
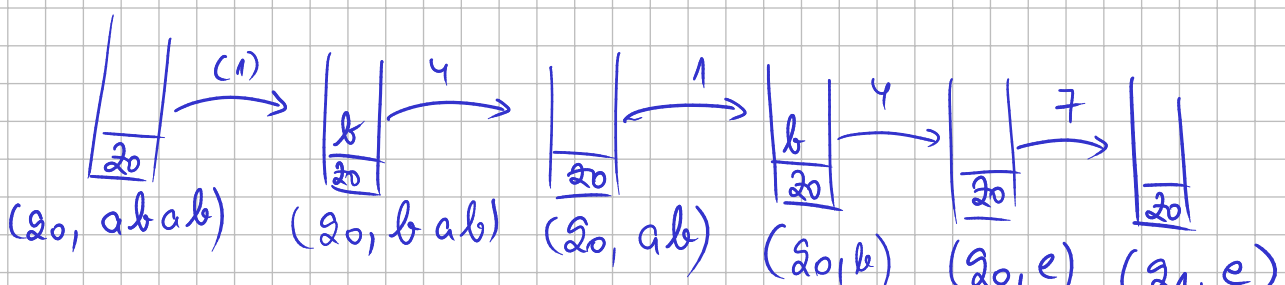
Propuneți un AS pentru limbajul:

$L_1 = \{ w \in \{a,b\}^* \mid \#a(w) = \#b(w) \}$

ASN



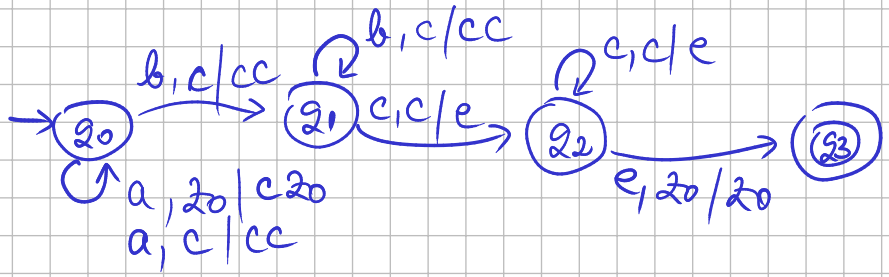
abab



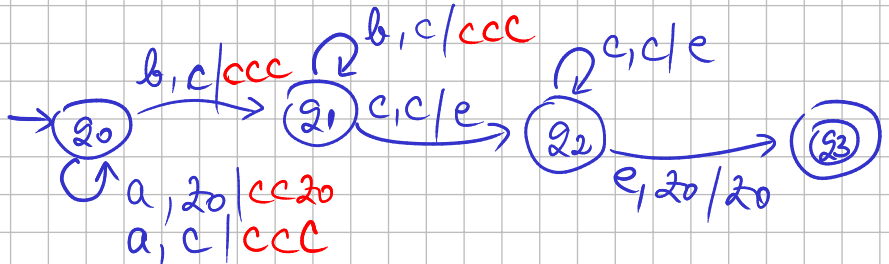
$L_2 = \{ a^i b^j c^k \mid i,j,k > 0,\ i+j=k \}$

$a^i b^j c^j c^i$

ASD



$L_3 = \{ a^n b^m c^{2(n+m)} \mid n, m \geq 1 \}$

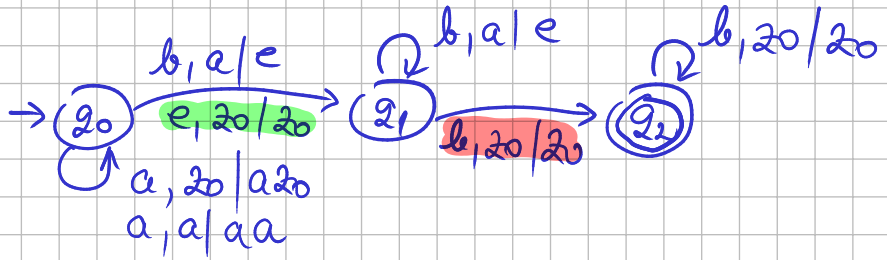


$L_4 = \{ a^n b^m \mid n < m \}$

$n < m \Rightarrow \exists\, k > 0$ a.î. $m + k = m$

$a^n b^m = a^n b^{n+k} = a^n b^n b^k$, $n \geq 0$, $k > 0$

ASN



$L_5 = \{ a^n b^m \mid n \neq m \} = \{ a^n b^m \mid n < m \} \cup \{ a^n b^m \mid m < n \}$

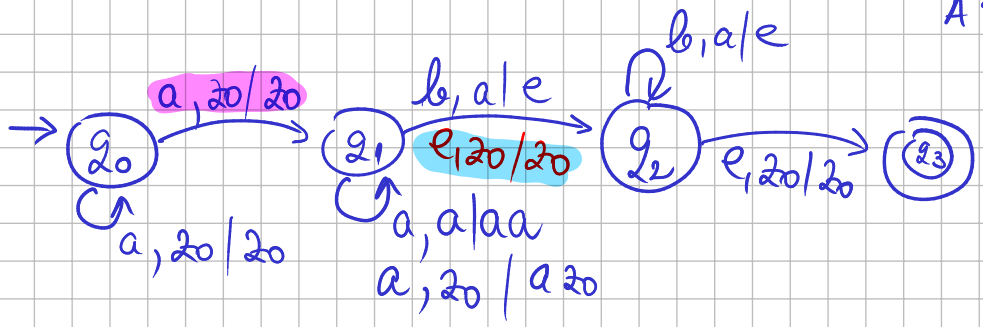
L_4 L_5'

AS pt. L_5'

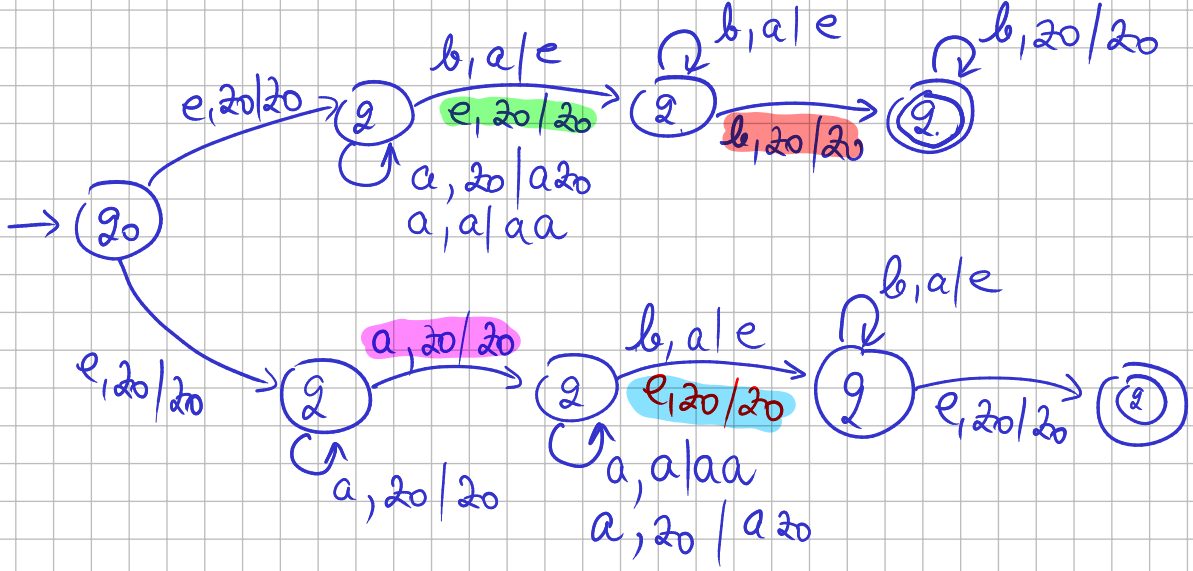
$m < n \Rightarrow \exists\, k > 0$ a.î. $n = m + k$

$a^n b^m = a^{m+k} b^m = a^k a^m b^m$, $k > 0$, $m \geq 0$

ASN

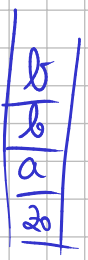


AS pt $L_5 = L_4 \cup L_5'$

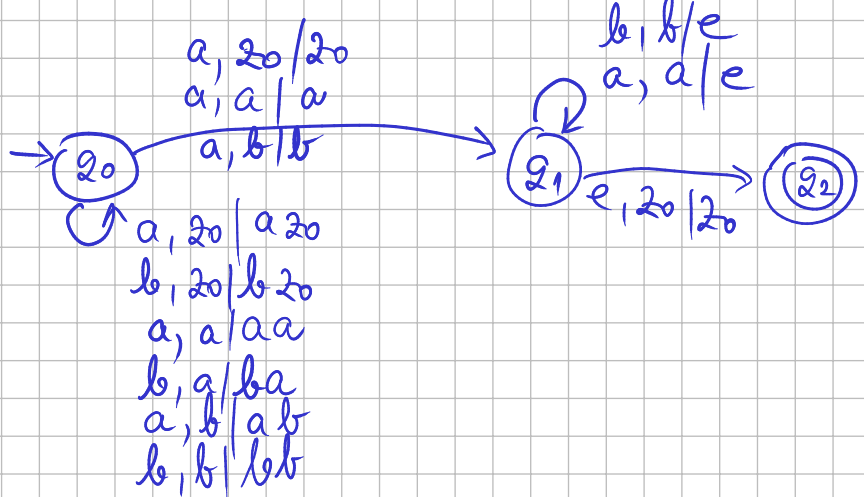


$L_6 = \{ w a w^R / w \in \{a,b\}^* \}$

$w = abb$



ASN



Construcția unui AS pornind de la G.I.C

Fie $G = (N, \Sigma, P, S)$. Vom ajunge la AS care acceptă $L(G)$

$AS = (\{q_0\}, \Sigma, \Sigma \cup N, m, q_0, S, \phi)$ – acceptare prin stivă vidă.

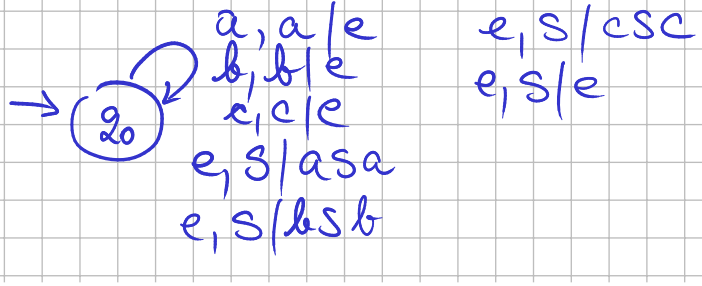
1) pentru orice $x \in \Sigma$ avem $m(q_0, x, x) = \{(q_0, e)\}$

2) pentru fiecare regulă de producție $p \in P (p \equiv A \to \alpha)$ avem : $(q_0, \alpha) \in m(q_0, e, A)$

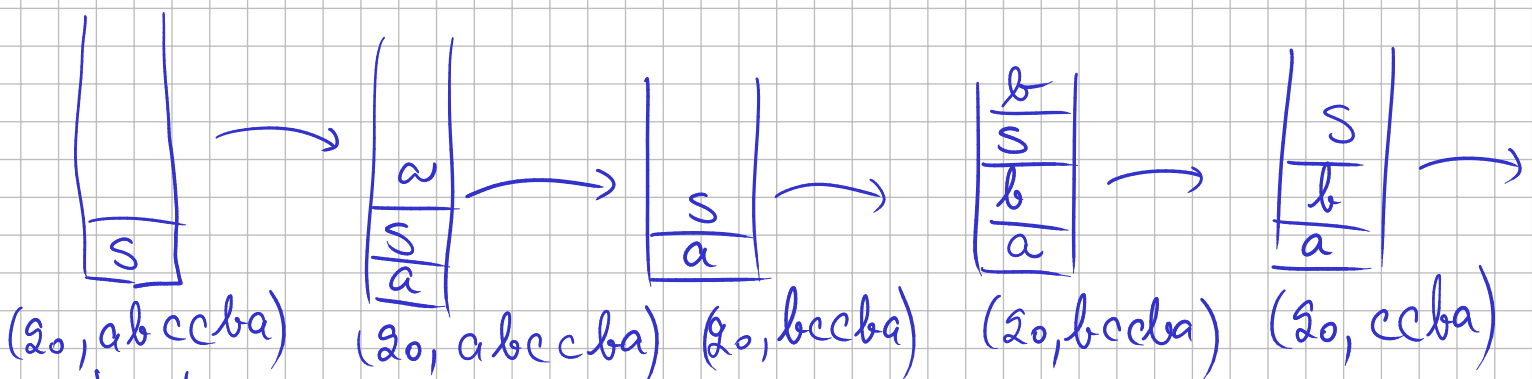
Determinați AS pornind de la G.I.C. pt:

$L_1 = \{ w w^R / w \in \{a,b,c\}^* \}$

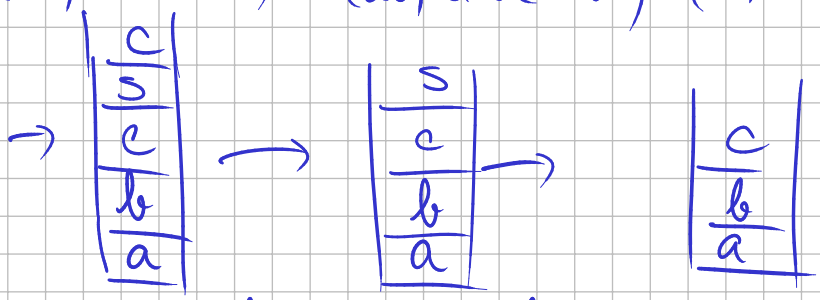
$S \to aSa / bSb / cSc / e$



abccba



$(q_0, abccba)$ $(q_0, abccba)$ $(q_0, bccba)$ $(q_0, bccba)$ $(q_0, ccba)$



$(q_0, ccba)$ (q_0, cba) (q_0, cba)

$$L_2 = \{ w \in \{a,b,c\}^* \mid \#_a(w) = \#_b(w) \}$$

$$S \to aSbS \mid bSaS \mid cS \mid e$$

